Individual Reflection – NewsBot Intelligence System

ITAI 2373 – Natural Language Processing

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Project: Midterm NewsBot Intelligence System

1. Which NLP techniques did you find most challenging? Which were most useful?

The most challenging part was combining different NLP techniques and getting the machine learning models to work correctly. Sometimes I received errors because the data did not match what the model expected.

The most useful techniques were text classification and Named Entity Recognition. I liked seeing how the system could identify news categories and extract information such as names, organizations, and locations.

2. How did you handle combining multiple NLP tasks?

I worked on the project one section at a time, starting with text preprocessing and moving through feature extraction, sentiment analysis, classification, and Named Entity Recognition.

After completing each section, I combined them into the NewsBot system. When I encountered errors, I reviewed the code and tested it again until it worked.

3. What real-world problems could this system solve?

I think NewsBot could help news organizations automatically organize large numbers of articles. It could also help businesses follow news about companies, products, and important events.

Sentiment analysis could help identify the tone of news coverage, while Named Entity Recognition could help find important names and organizations. This could save time and make information easier to find.

4. What are the potential risks of automated news analysis?

One risk is that the system can make incorrect predictions. During testing, the climate change article was classified as Business with low confidence, and its sentiment was labeled positive even though the article discussed serious concerns.

This showed me that AI does not always understand the full meaning of text. Bias in training data could also affect results. I believe human review is important when using automated news analysis.

5. What NLP topics are you most excited to explore next?

I would like to learn more about topic modeling and advanced language models such as BERT.

Topic modeling interests me because it can help discover common subjects in large collections of documents. I would also like to understand how newer language models can improve classification and text analysis.

6. How did you divide the work and ensure quality?

I completed the project independently, so I was responsible for the coding, testing, analysis, and documentation.

I organized the work by following the notebook sections and completing them one at a time. I checked the code outputs and tested the final system using three different articles.

This helped me practice solving problems and checking the quality of my work.

7. How will you present this project to potential employers?

I would include NewsBot in my GitHub portfolio to show the NLP and machine learning skills I practiced.

I would explain how the system classifies news articles, analyzes sentiment, and extracts important information. I would also show the model comparison results, including the Naive Bayes model that achieved 93.75% accuracy.

Final Reflection

Overall, this project was challenging, but it helped me understand how different NLP techniques can work together.

I learned that writing code is only part of building an AI system. Understanding the results, finding errors, and recognizing limitations are also important.

This project helped me become more comfortable working with Python, NLP libraries, and machine learning models. It also gave me useful experience that I can apply to future projects.